A second failure of the thinness assertion

Human Compression project

Abstract

The thinness clause for prime $q-1>7$ is separate from the clause for $q=8$. We settle it by an explicit example in Sz(32), using the same coset argument as for the previously verified Sz(8) example. Its parabolics have orders $62, 62, 82$, and each actual panel has three elements although the corresponding algebraic residue has two.

Work in $F=\mathbb{F}_2[X]/(X^5+X^2+1)$. Integers from zero to31 encode coefficients in the basis $1, X, X^2, X^3, X^4$; field addition is exclusive-or. The independent arithmetic reader checks all31 nonzero polynomial classes have inverses. In the classical Suzuki model put

$$a=\begin{pmatrix}0&0&0&1\\0&0&1&0\\0&1&0&0\\1&0&0&0\end{pmatrix},\quad b=\begin{pmatrix}12&3&0&1\\3&0&1&0\\20&4&0&3\\30&20&3&12\end{pmatrix},\quad c=\begin{pmatrix}26&5&0&1\\5&0&1&0\\29&16&0&5\\19&29&5&26\end{pmatrix}.$$

Their membership in the ambient group is explicit. With $w=a$ and $z(t)=u(0,t)$, where

$$u(s,t)=\begin{pmatrix}1&0&0&0\\s&1&0&0\\s^9+t&s^8&1&0\\s^{10}+st+t^8&t&s&1\end{pmatrix},$$

we have $b=z(3)wz(3)$ and $c=u(1,5)wz(1)wu(1,5)^{-1}$. The ambient matrix construction and maximality of the full split normalizer D_{62} in Sz(32) remain explicit classical premises.

Literal polynomial multiplication verifies that a, b, c are distinct nonidentity involutions and that

$$|ab|=31,\qquad |ac|=31,\qquad |bc|=41,\qquad (ac)^{15}(ab)^{15}a=(bc)^{20}. \tag{1}$$

Define $G_0=\langle a,b\rangle$, $G_1=\langle a,c\rangle$ and $G_2=\langle b,c\rangle$. Their respective orders are $62, 62, 82$. Coprimality gives $G_0\cap G_2=\langle b\rangle$ and $G_1\cap G_2=\langle c\rangle$. The subgroup $G_0\cap G_1$ contains $\langle a\rangle$, which is maximal in the prime dihedral group G_0. If the intersection were larger, then $G_0=G_1$, placing b, c in a group of order62 and contradicting $|bc|=41$. Thus $G_0\cap G_1=\langle a\rangle$, and the triple intersection is trivial.

Theorem 1. *The rank-three coset geometry $\Gamma(\mathrm{Sz}(32); G_0, G_1, G_2)$ is RWPRI and satisfies the literal algebraic local 2-transitivity condition of L114. It is not thin and is not flag transitive.*

Proof. An incident pair of left cosets has a common representative and hence extends to a chamber. Thus the pregeometry is a geometry. For the empty subset in the RWPRI definition, G_0 is maximal in $G=\mathrm{Sz}(32)$. Each pair intersection has prime index31 or41 in the corresponding parabolic, and the trivial triple intersection is maximal in each pair intersection. These verify every required subset. Each source algebraic rank-one residue is the regular two-point action $C_2/1$, so is 2-transitive.

The base edge (G_1, G_2) has the standard extensions G_0, cG_0. Put $x=(ac)^{15}$. The coset xG_0 meets G_1 at x and, by (1), meets G_2 at $(bc)^{20}$. It differs from G_0, since x has order31

and $G_0 \cap G_1 = \langle a \rangle$. If $xG_0 = cG_0$, then $cx \in \langle a \rangle$. The alternative $cx = 1$ contradicts the orders of c, x. The alternative $cx = a$ gives $(ac)^{15} = ca = (ac)^{30}$, contradicting $|ac| = 31$. We have found a third chamber, so the geometry is not thin. This chamber has no common representative: such a representative would lie in $G_1 \cap G_2 = \{1, c\}$ and identify its third coset with one of the standard cosets. It therefore cannot be a translate of the base chamber, proving failure of flag transitivity. □

The independent reader checks every product and inverse in the three returned subgroup lists and verifies generation, intersections and all cosets in each panel. It obtains exactly three cosets in each case, with the following representatives:

missing type	standard representatives	extra representative
0	$1, c$	$(ac)^{15}$
1	$1, b$	$(ab)^{15}$
2	$1, a$	$(ba)^{15}$

The reader explicitly matches each displayed representative with its complete returned coset. Thus the table is certified along with (1).

Source scope and attribution. The source is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* Sz(q), Note di Matematica 19(1) (1999), 43–63. Its definitions on printed pp. 45–46 use algebraic residues formed from parabolic intersections, and count actual chambers for thinness. The example has $q - 1 = 31 > 7$ prime and therefore refutes the thinness clause in the second branch of Theorem 6.1 on printed p. 61. The previously accepted rank bound and rank-two count in that branch are unaffected. A version requiring flag transitivity or local 2-transitivity on full actual panels has different hypotheses.

The field and split pair were retained from earlier work. The new contribution is this third involution, its short coset identity and the exact source application, reusing the preceding Sz(8) argument. The bounded discovery stopped after156 prescribed candidates. Its unaltered output is retained; the independent reader verifies this witness alone and makes no classification claim. No new original, principal-refutation count, open-conjecture, Lean or compression credit is assigned. Distinct mathematical review is required before adoption.